

# Laksh's Daily Brief: July 26, 2026

## 1. Top World, AI, Tech, and Market News

### 1. Nvidia Locks Down SK Hynix Memory in \$500B+ AI Deal

- **What happened:** Nvidia and South Korea's SK Group announced a \$500 billion+ AI initiative, including a long-term high-bandwidth memory (HBM) supply agreement with SK Hynix ([CNBC](#), [Reuters](#)).
- **Why it matters:** HBM is the critical memory component for AI GPUs. Nvidia is securing supply ahead of a global HBM shortage, ensuring its ability to meet surging demand for AI training and inference. SK Hynix benefits from guaranteed revenue; rivals like Micron and Samsung may face supply constraints.
- **Second-order effects:** The deal also includes large-scale AI data center construction, boosting demand for power, cooling, and networking equipment. It signals that AI infrastructure investment is shifting from "if" to "how fast."

### 2. TSMC Expands Arizona Investment by \$100B to \$265B Total

- **What happened:** TSMC raised its planned U.S. investment by \$100 billion, bringing total Arizona commitments to \$265 billion — the largest foreign direct investment in U.S. history ([MarketWise](#)).
- **Why it matters:** This accelerates the U.S. push for semiconductor self-sufficiency. TSMC's advanced manufacturing (including CoWoS packaging) is now largely booked through 2027. However, pressure from President Trump to shift production to the U.S. is hitting TSMC's margins ([CNBC](#)).
- **Who benefits:** U.S. tech companies that rely on advanced chips (Nvidia, AMD, Apple) get supply chain diversification. Construction and equipment suppliers (e.g., ASML, Applied Materials) also benefit.

### 3. Google's SpaceX Stake Disclosed at \$94.1B — 6% Ownership

- **What happened:** Google disclosed a \$94.1 billion stake in SpaceX, representing a 6% ownership position ([WSJ](#)).
- **Why it matters:** This reveals the deep strategic alignment between cloud computing and space infrastructure. SpaceX's Starlink provides low-latency connectivity for remote data centers and edge computing, while Google's cloud can offer AI services in space. It also underscores the growing value of private space companies ahead of a potential SpaceX IPO.

### 4. Alphabet's Capex Forecast Surges, Stock Craters

- **What happened:** Alphabet reported strong cloud revenue growth but increased its capex forecast again, spooking investors who worry about spending without commensurate returns ([Reuters](#)). The stock fell ~7% in the watchlist snapshot.
- **Why it matters:** The market is punishing companies that invest heavily in AI infrastructure without clear profitability timelines. Alphabet's free cash flow is under pressure as capex rises ([Reuters](#)). This sets the stage for a crucial week ahead as Amazon, Meta, and Microsoft report earnings.

### 5. Apple Emerges as the "Anti-Capex" Hedge

- **What happened:** Analysts are positioning Apple as a Magnificent Seven hedge against hyperscaler spending, because Apple's AI strategy relies less on massive data center buildout and more on device-level AI ([CNBC](#)).
- **Why it matters:** If investors rotate away from high-capex tech stocks (Alphabet, Amazon, Meta, Microsoft) toward lower-capex ones (Apple, possibly others), it could reshape sector dynamics. Apple's model suggests that edge AI (on-device) may be more capital-efficient than cloud AI.

## 6. OpenAI Signs Open-Weights Letter, Coalition Grows to 35

- **What happened:** OpenAI signed the "Open Weights and American AI Leadership" letter that it had initially skipped, joining a coalition of 35 companies including Anthropic, Google, and Meta ([FourWeekMBA](#)).
- **Why it matters:** The shift toward open-weight AI models (where model parameters are publicly released) changes the competitive landscape. It could accelerate AI adoption by lowering barriers for startups and researchers, but also raises concerns about misuse and diminished commercial moats for AI labs.

## 7. Anthropic Releases Claude Opus 5 — Cheaper, Stronger

- **What happened:** Anthropic launched Claude Opus 5, which approaches the capabilities of its flagship Claude Fable 5 at half the price ([CNET](#)).
- **Why it matters:** This is a classic "more for less" AI model release, putting pressure on OpenAI and Google to lower prices or improve performance. It accelerates the commoditization of AI models, benefiting consumers but potentially squeezing margins for AI labs.

## 8. Ukraine Strikes Iranian Vessels in Caspian Sea

- **What happened:** Ukraine struck Iranian vessels in the Caspian Sea, and Iran accused Kyiv of a "hostile and criminal act" ([CNBC](#)). Simultaneously, Saudi Arabia struck Houthi targets in Yemen.
- **Why it matters:** The war is widening geographically. Any disruption to the Strait of Hormuz or Caspian Sea shipping could spike oil prices further. Crude is already near \$100/barrel, driving up inflation expectations and rate hike odds. This is a direct risk for supply chains and energy costs.

## 9. Fed Rate Hike Odds Surge as Oil Rips Higher

- **What happened:** Fed funds futures are pricing in an 82% probability of a rate hike at the September meeting, driven by rising oil prices and inflation fears ([CNBC](#)). The July meeting (this week) is expected to hold rates steady, but the tone will be critical ([CBS News](#)).
- **Why it matters:** Higher rates make future cash flows (especially for high-growth tech) less valuable and increase borrowing costs for companies with heavy capex plans. This is a headwind for AI infrastructure stocks.

## 10. EU Imposes 21st Russia Sanctions Package, Including Crypto Ban

- **What happened:** The EU agreed on a new sanctions package against Russia, including a price cap on Russian oil, a crypto ban, and shadow fleet seizures ([DW](#), [Kyiv Independent](#)).

- **Why it matters:** This tightens the squeeze on Russia's war funding. The oil price cap could further disrupt global oil markets, while the crypto ban aims to close loopholes used for sanctions evasion. It may also increase volatility in crypto markets.

## 11. Data Centers on Track for 20% of U.S. Power by 2035

- **What happened:** A Bloomberg analysis cited by Yahoo Finance projects that data centers will consume ~20% of U.S. electricity by 2035, up from 5.9% today ([Yahoo Finance](#)).
- **Why it matters:** This staggering growth is straining the grid, raising costs, and sparking local opposition (e.g., North Carolina residents pushing to stay data-center-free ([The Cool Down](#))). It also drives demand for new power sources, including nuclear ([Forbes](#)). Expect higher utility bills and regulatory battles.

## 12. Stock Market "Triple Threat" — Yields, Oil, Dollar

- **What happened:** Treasury yields, oil prices, and the U.S. dollar all surged, creating a "technical triple threat" that pushed the S&P 500 below key chart support ([MarketWatch](#)).
- **Why it matters:** Rising yields make bonds more attractive vs. stocks; high oil hurts corporate margins; a strong dollar pressures multinational earnings. This is a broad headwind, especially for growth and tech stocks. The market is now heavily dependent on this week's mega-cap earnings and Fed meeting to reverse the slide.

## 2. Infrastructure and Supply-Chain Logic

### 2.1 AI Demand → Cloud Capex → GPUs → HBM → Networking → Power

The Nvidia-SK Hynix deal is a direct link in this chain:

- **AI demand** from enterprises and consumers drives cloud providers (Google, Amazon, Microsoft, Meta) to build more data centers.
- **Cloud capex** is rising sharply; Alphabet just raised its forecast again. This money flows to **GPUs** (Nvidia, AMD) and **accelerators** (Broadcom, Marvell).
- **GPUs** require **HBM** (high-bandwidth memory). Nvidia secured SK Hynix supply, but the global HBM shortage means other GPU makers (AMD, Intel) may struggle. This also benefits **HBM equipment makers** and **packaging** companies (e.g., TSMC's CoWoS).
- **Networking** (optical transceivers, switches) scales with data center size. Companies like **Marvell, Coherent, and Lumentum** benefit.
- **Power** is the ultimate bottleneck. The projected 20% of U.S. electricity by 2035 means **utilities** (Constellation Energy, Vistra, NextEra) and **nuclear** developers are critical. However, grid buildout costs are soaring ([WSJ](#)).

### 2.2 TSMC's U.S. Expansion: Geopolitics Meets Margins

TSMC's \$100B Arizona boost is a political and economic story:

- **Geopolitical rationale:** Reduce reliance on Taiwan for advanced chips, given China's tensions. The U.S. wants to secure its AI supply chain.
- **Margin impact:** Building fabs in the U.S. is more expensive than in Taiwan (higher labor, construction, and regulatory costs). TSMC's margins are being squeezed, as noted in the CNBC piece ([CNBC](#)). Investors should watch for potential margin erosion in future earnings.
- **Second-order effect:** Higher chip costs may be passed down to Nvidia, AMD, and Apple, potentially raising AI infrastructure costs for everyone.

### 2.3 The Nuclear Option: AI's Power Problem

Several sources point to nuclear as a solution to AI's power demand ([Forbes](#), [247wallst](#)). However, the Bulletin of the Atomic Scientists warns that next-gen nuclear is years away and may be hype ([The Bulletin](#)). The tension is:

- **Short-term:** Natural gas and renewables will fill the gap, but gas prices are rising and renewables are intermittent.
- **Long-term:** Small modular reactors (SMRs) could be the answer, but they face regulatory, cost, and timeline hurdles. Utilities with existing nuclear (e.g., CEG, VST) are positioned to benefit from power purchase agreements with hyperscalers.

## 3. Watchlist: Earnings, Guidance, and Notable Movers

### Earnings in the Last 24 Hours (from sources)

- **No major watchlist earnings reported in the last 24 hours** from the source bundle. However, Alphabet (GOOGL) reported on July 22, and its stock dropped ~7% in the watchlist snapshot. Amazon, Meta, Microsoft report this week (July 29-30).

### Notable Movers (>3% or major news)

(Price moves based on the watchlist snapshot, approximate and possibly delayed)

| Ticker | Price    | Move           | Reason (from sources)                                                                                          |
|--------|----------|----------------|----------------------------------------------------------------------------------------------------------------|
| TSLA   | \$319.69 | <b>-14.52%</b> | No specific news in sources; likely continued selloff after weak Q2 delivery figures or broader tech weakness. |
| GOOGL  | \$317.69 | <b>-7.13%</b>  | Alphabet's capex forecast increase and mixed market reaction to earnings ( <a href="#">Reuters</a> ).          |
| CMCSA  | \$21.92  | <b>-6.80%</b>  | No specific source; may be reacting to broader market or sector rotation away from media.                      |

| Ticker | Price    | Move    | Reason (from sources)                                                                    |
|--------|----------|---------|------------------------------------------------------------------------------------------|
| AMZN   | \$233.66 | -4.57%  | General tech weakness ahead of its own earnings next week.                               |
| META   | \$606.1  | -3.36%  | Same as AMZN; market bracing for its capex commentary.                                   |
| CRDO   | \$236.5  | +3.61%  | Likely benefiting from AI infrastructure optimism (optical networking for data centers). |
| MU     | \$990.21 | +3.20%  | Positive on Nvidia's HBM supply deal with SK Hynix; Micron is also an HBM player.        |
| RDW    | \$9.28   | +3.23%  | No specific source; may be tied to space/nuclear interest.                               |
| IMAX   | \$43.96  | +11.86% | No specific source; possibly a movie release or earnings beat not covered in bundle.     |
| INTC   | \$100.23 | -2.33%  | Continued pressure from competitive position in AI chips.                                |
| AMD    | \$539.69 | -2.29%  | Underperformance vs. Nvidia; HBM supply concerns may hurt AMD's GPU plans.               |

### Key Numbers to Watch

- **Alphabet:** Q2 cloud revenue beat expectations, but capex forecast raised — the market focused on the latter.
- **TSLA:** No earnings in the last 24 hours; the ~14% drop is likely a continuation of prior week's selloff (not covered in sources, but watchlist shows it).
- **Nvidia:** No earnings, but the SK Hynix deal is a major positive catalyst.

## 4. Beginner Knowledge Lens

1. **Capex vs. Free Cash Flow:** Capital expenditure (capex) is money spent on long-term assets (data centers, chips). Free cash flow (FCF) is the cash left after capex. If capex grows faster than revenue, FCF shrinks. Investors worry when FCF drops because it limits dividends, buybacks, and debt repayment. Google's rising capex is squeezing its FCF.

2. **Oil Prices and Inflation:** Oil is a key input for many goods (transportation, plastics). When oil prices rise, inflation expectations rise. The Fed then may raise interest rates to cool the economy. Higher rates make stocks (especially growth stocks) less attractive because future earnings are discounted more heavily.
3. **Open-Weight vs. Closed-Weight AI Models:** Open-weight models release the trained parameters (the "weights") publicly. Anyone can use them, modify them, or run them. Closed-weight models (like earlier versions of GPT-4) keep the weights secret. Open models reduce barriers to entry but may also reduce the commercial advantage of AI labs.
4. **Data Center Power Consumption:** A typical hyperscale data center can use 100-200 megawatts (MW) of electricity — enough to power 50,000-100,000 homes. The projected growth to 20% of U.S. electricity by 2035 means every new data center must compete with residential and industrial users for power, driving up costs and leading to local opposition.
5. **HBM (High-Bandwidth Memory):** HBM is a type of DRAM that is stacked vertically and placed close to the GPU. It provides extremely fast data transfer (up to ~1 TB/s in HBM3e). It's essential for AI training because models need to read and write data quickly. The global HBM market is dominated by SK Hynix, Samsung, and Micron.
6. **Technical Triple Threat:** When Treasury yields, oil prices, and the U.S. dollar all rise at the same time, it's historically been a tough environment for stocks. Yields compete with stocks for investor money, oil hurts corporate profits, and a strong dollar reduces the value of overseas earnings for U.S. companies.

## 5. Terms Used Today

1. **Free Cash Flow (FCF):** Cash generated from operations minus capital expenditures. *Example: If a company earns \$100B in cash from operations and spends \$60B on capex, its FCF is \$40B. A declining FCF signals that growth investments are consuming cash.*
2. **HBM (High-Bandwidth Memory):** A 3D-stacked DRAM technology that provides high-speed data transfer for AI GPUs. *Rough benchmark: HBM3e bandwidth ~1 TB/s; typical DDR5 DIMM bandwidth ~0.05 TB/s. HBM is essential for training large models.*
3. **Capex (Capital Expenditure):** Money spent on physical assets like buildings, machinery, or data centers. *Example: Google's \$50B+ in AI capex this year is its expenditure on data centers, GPUs, and networking equipment.*
4. **Hyperscaler:** A large cloud provider that operates massive data centers at scale — e.g., Amazon Web Services (AWS), Microsoft Azure, Google Cloud. They are the primary customers of AI hardware.
5. **Open-Weight Model:** An AI model whose trained parameters (weights) are publicly released, allowing anyone to download, run, and modify it. *Example: Meta's Llama models are open-weight; early GPT-4 was not.*
6. **CoWoS (Chip-on-Wafer-on-Substrate):** A 2.5D packaging technology used by TSMC to stack multiple chips (e.g., GPU + HBM) on a single substrate. It's a bottleneck for AI chip production because capacity is limited.

7. **Shadow Fleet:** Ships used to transport oil or goods in violation of sanctions, often by using opaque ownership, fake flags, or disabling tracking. The EU's new sanctions target these vessels.
8. **Strait of Hormuz:** A narrow waterway between Iran and Oman through which about 20% of the world's oil passes. Any disruption there (e.g., from war) can spike oil prices globally.
9. **P/E Ratio (Price-to-Earnings):** A valuation metric: stock price divided by earnings per share.  
*Example: A stock at \$100 with \$5 EPS has a P/E of 20. A high P/E (e.g., 30+) often implies high growth expectations, which are vulnerable to rate hikes or earnings misses.*
10. **Grid Interconnection:** The process of connecting a new power plant or data center to the electrical grid. It can take years due to permitting and equipment backlogs, pushing up costs ([WSJ](#)).

## 6. Sources Pulled

### AI & Tech Industry

- [CNBC — Nvidia locks down memory supply from SK Hynix as part of \\$500 billion AI deal](#)
- [Reuters — Nvidia, SK Group unveil \\$500 billion-plus AI data centers initiative, memory partnership](#)
- [FourWeekMBA — OpenAI Signs the Open-Weights Letter It Skipped at Launch](#)
- [CNET — Anthropic Releases Claude Opus 5](#)
- [WSJ — Google Discloses \\$94.1B in SpaceX Stock, Marking 6% Stake](#)
- [CNBC — Apple becomes the Mag7 hedge against hyperscaler spending](#)

### Semiconductor Manufacturing & Supply Chain

- [MarketWise — TSMC Boosts U.S. Chip Production by \\$100 Billion](#)
- [CNBC — Trump's push for American-made AI chips hits TSMC's margins](#)
- [Tom's Hardware — Zeiss expands German site that caps ASML's EUV scanner output](#)
- [Tom's Hardware — Chinese CXMT DRAM doesn't look like the budget savior](#)

### Data Centers, Power, & Infrastructure

- [Yahoo Finance — Data Centers on Track to Suck Up a Fifth of US Power Use by 2035](#)
- [WSJ — America's Demand for Power Is Soaring—and So Are the Costs of Building Out the Grid](#)
- [247wallst — Nuclear Could Solve America's Looming Natural Gas Crisis — Except For This 1 Thing](#)
- [Forbes — The AI Boom Is Making Nuclear Power Bankable Again](#)
- [The Bulletin — Data centers powered by next-gen nuclear? Don't fall for Big Tech's PR hype](#)
- [Energy News Beat — Over 100°F Heat Adding Stress to the AI-laden Grid](#)
- [The Cool Down — North Carolina residents push to stay data center-free](#)

### Markets, Fed, & Macro

- [CNBC — Odds of Federal Reserve rate hike surge as oil prices rip higher](#)
- [CBS News — Will the Federal Reserve raise interest rates? Here is what experts predict](#)
- [USA Today — Fed chair 'an enigma': Why July interest rate move is hard to predict](#)
- [MarketWatch — The Treasury market is sending Fed Chair Kevin Warsh a clear warning](#)
- [MarketWatch

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